Ex.No : 01 IMPLEMENTATION OF METADATA EXTRACTION FROM A DIGITAL IMAGE

Date :

AIM:

To extract, examine, and analyze metadata from a digital image using a forensic tool and interpret its significance in digital forensic investigations.

SOFTWARE/EQUIPMENTS REQUIRED:

- Windows 10/11 or Linux (Ubuntu/Kali)
- ExifTool

THEORY:

Metadata is commonly referred to as "data about data." It is descriptive information stored within or associated with a digital file that provides details about the file's properties, origin, creation, and modification history. Although metadata is typically invisible during normal file viewing, it can be extracted using specialized forensic tools. In digital forensics, metadata plays a crucial role in investigating digital evidence. It helps investigators determine the origin, authenticity, and history of a file, reconstruct event timelines, verify evidence integrity, and identify possible signs of tampering or manipulation.

TOOL USED:

ExifTool is a command-line tool used to read, write, edit, and extract metadata from digital files such as images, videos, audio, and documents. It is widely used in digital forensics to analyze file metadata.

ALGORITHM:

1. Download and install ExifTool.
2. Open Command Prompt (Windows) or Terminal (Linux).
3. Navigate to the folder containing the image.
4. Execute the following command to extract the metadata.

   Example: exiftool image1.jpg
5. Observe the metadata displayed by ExifTool.
6. Record important metadata fields.
   - File Name, File Size
   - File Type
   - MIME Type
   - Image Width
   - Image Height
   - Date Created, Date Modified
   - Camera Make, Camera Model
7. Interpret the extracted metadata and document your observations.
8. End

PROCEDURE

Step 1: Download ExifTool

1. Dowload ExifTool by Phil Harvey.
2. Find the Windows Executable download.
3. Download the Windows ZIP file.
4. Extract the downloaded ZIP file.
5. Locate the executable file, usually named exiftool(-k).exe.
6. Rename exiftool(-k).exe to exiftool.exe.
7. Keep the executable in a simple folder, such as D:\DMF\ExifTool.

Step 2: Prepare the experiment folder

1. Open File Explorer.
2. Create the folder D:\DMF\Experiment1.
3. Copy one sample JPEG image into this folder.
4. Rename the image to image1.jpg.
5. If the extracted package contains an exiftool_files folder, keep that folder alongside exiftool.exe.

Step 3: Open Command Prompt

1. Press Windows + R.
2. Type cmd.
3. Press Enter.

Step 4: Navigate to the experiment folder

1. Type this command and press Enter:

```bat
cd /d D:\DMF\Experiment1
```

Step 5: Extract the image metadata

```bat
exiftool.exe image1.jpg
```

Sample Output:

```
File Name                       : image1.jpg
Directory                       : D:/DMF/Experiment1
File Size                       : 245 kB
File Type                       : JPEG
MIME Type                       : image/jpeg
Image Width                     : 1920
Image Height                    : 1080
Image Size                      : 1920x1080
```

Output

S.No.	Item	Description